GOVT 6019: Homework 8 — Linear regression I

Fall 2026

Melissa Sands (Instructor) Izza Malik (Teaching Assistant)

Question 1: True or false

State whether each claim is true or false. Give a one-sentence reason.

a. R^2 measures how steep the regression line is.

b. If every residual is zero, then $R^2 = 1$.

c. Changing X from dollars to thousands of dollars changes R^2.

d. Adding 10 to every value of Y changes the OLS slope.

e. The OLS residuals average to zero within each value of X.

f. If $R^2 = 0.25$, the line's predictions are off by 75% on average.

g. The OLS line always passes through the point $(\bar{X}, \bar{Y})$.

h. The OLS intercept is always the predicted value for some real unit in the data.

i. Regressing Y on X and regressing X on Y give the same slope.

j. A slope of $\hat{\beta}_1 = 2$ means that increasing X by one unit causes Y to rise by 2 units.

Question 2: Knocking on doors

Before a local election, a county party sent door-knocking volunteers into six precincts. For each precinct, X is the number of volunteers per 1,000 residents and Y is turnout, the percentage of registered voters who voted.

Precinct	X	Y
1	0	42
2	2	56
3	4	59
4	4	60
5	6	58
6	8	61

Do parts (a)–(f) "by hand" (with a calculator). Show your work.

a. Compute $\bar{X}$ and $\bar{Y}$.

b. Make a table with columns $X_i - \bar{X}$, $Y_i - \bar{Y}$, $(X_i - \bar{X})(Y_i - \bar{Y})$ and $(X_i - \bar{X})^2$, and the sum of each column. Use it to compute $\widehat{\text{Cov}}(X, Y)$ and s_X^2.

c. Compute the OLS slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$. Write the fitted line.

d. Compute the fitted value $\hat{Y}_i$ and the residual $\hat{\varepsilon}_i$ for each precinct. Check that $\sum_i \hat{\varepsilon}_i = 0$ and $\sum_i X_i \hat{\varepsilon}_i = 0$.

e. Compute TSS, ESS and SSR. Check that TSS = ESS + SSR. Compute R^2.

f. Compute s_Y and the correlation $\hat{r}$. Check that $R^2 = \hat{r}^{\,2}$.

g. Interpret $\hat{\beta}_1$ and $\hat{\beta}_0$ in words, in the units of X and Y.

h. The party chair concludes: "Every extra volunteer per 1,000 residents raises turnout by 2 points." Does the regression show this? Explain.

i. Enter the data in R and check your answers to (c) and (e) with `lm()` and `summary()`.

Question 3: Party and the importance of gay rights

The 2012 VOTER Study asked respondents how much they agree that "Gay rights are an important issue." The file `voter2012.csv` holds the answer in `important_gayr` and party identification in `pid3`. Its codebook is `voter2012_codebook.md`.

```
library(tidyverse)
voter <- read_csv("data/voter2012.csv")   # adjust to your folder layout
```

a. Set the "Don't know" answers to `important_gayr` to missing. Create a variable `dem` equal to 1 for Democrats, 0 for Republicans and missing for everyone else. How many respondents have a value for both `important_gayr` and `dem`?

b. Compute the mean of `important_gayr` among Democrats and among Republicans, and the difference between the two.

c. Regress `important_gayr` on `dem` with `lm()`. Report the intercept, the slope and R^2. How do the intercept and slope compare with your answers to (b)?

d. Interpret the intercept and the slope in words. Do Democrats or Republicans rate gay rights as the more important issue?

e. Suppose you had instead created `rep`, equal to 1 for Republicans and 0 for Democrats. Without running it, what would the intercept and slope of `lm(important_gayr ~ rep)` be? Run it to check.

Question 4: Hours worked over a lifetime

The 2024 ANES asked respondents who had worked at some point in the past 12 months: "During the past 12 months, in the weeks worked, how many hours did you usually work each week?" The file `anes2024_work.csv` holds one row per respondent: an ID, the answer (`hours`) and the respondent's age (`age`). Empty cells are respondents who were not asked or who did not answer. Its codebook is `anes2024_work_codebook.md`.

```
library(tidyverse)
work <- read_csv("data/anes2024_work.csv") |>   # adjust to your folder layout
  drop_na()
```

a. How many respondents have both `hours` and `age`? Make a scatterplot of `hours` against `age`. Many respondents share the same values, so use `geom_jitter()` with a small `alpha`. Describe what you see.

b. The code below sorts respondents into age groups. Compute the mean of `hours` and the number of respondents in each group. Describe how mean hours change with age.

```
work <- work |>
  mutate(age_group = cut(age, breaks = c(17, 29, 39, 49, 59, 69, 80)))
```

c. Regress `hours` on `age` with `lm()`. Interpret the slope in words.

d. Interpret the intercept. Does it describe anyone in the data?

e. Use `summary()` to find R^2 and the residual standard error. Interpret each in words.

f. Use the fitted line to predict weekly hours for a 25-year-old and for a 75-year-old. Compare each prediction with the mean for that respondent's age group from (b).

g. A researcher sees a slope of -0.145 and an R^2 of 0.021, and concludes: "Age has almost nothing to do with how many hours people work." Do you agree? Use your answers to (b) and (f).

h. Refit the regression without the answers above 100 hours. Describe how the slope and R^2 change.